

RAVINDU WEERAKOON

BSc Hons in Computer Science
Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate at the University of Peradeniya with practical experience in machine learning, cyber-security, and full-stack development. Worked on projects involving behavioral biometrics, deep learning, computer vision, and embedded systems using modern frameworks and tools. Interested in building intelligent, secure, and scalable software systems through research and engineering.

EDUCATION

Bachelor of Science in Computer Science (Honors)

University of Peradeniya

Expected Graduation: 2026

Peradeniya, Sri Lanka

- **Relevant Coursework:** Artificial Intelligence, Neural Networks and Deep Learning, Natural Language Processing, Machine Learning, Digital Image Processing, Computer Graphics Programming, Information and Network Security, Data Structures, Database Management Systems, Object-Oriented Analysis and Design, Software Engineering, Computational Mathematics

School Education

G.C.E. Advanced Level & Ordinary Level Examinations

Completed: 2020

Sri Lanka

- **G.C.E. Advanced Level (A/L):** 1A, 1B, 1C
Physical Science Stream (Z-Score: 1.4449)
- **G.C.E. Ordinary Level (O/L):** 8As, 1B

RESEARCH PROJECTS

Behavioral Biometric Authentication System

Jan 2026 – Present

- Engineered a privacy-preserving authentication system using Keystroke and Mouse Trajectory (KMT) dynamics, applying the Johnson-Lindenstrauss Lemma for secure dimensionality reduction.
- Designed a curriculum learning pipeline with Deep SVDD for anomaly detection, leveraging feature entropy and Optuna tuning to maximize statistical robustness and model convergence.

SOFTWARE & ENGINEERING PROJECTS

Machine Learning & Deep Learning

- **Biomedical Named Entity Recognition Pipeline** (*Python, PyTorch, Transformers*) **May 2026 – Present**
Developed a sequence-labeling pipeline benchmarking 5 embedding types (BERT, ELMo) and 3 tokenizers to execute token-level BIO-tag disease extraction via Bi-LSTM architectures.
- **Tea Leaf Disease Classifier** (*Python, PyTorch, OpenCV*) **Apr 2026 – May 2026**
Engineered a 4-stage progressive unfreezing CNN pipeline (ResNet50/VGG16) featuring custom Otsu's segmentation and an intelligent F1-score weighted Hybrid Ensemble that boosted diagnostic accuracy to 90.54%.
- **Twitter Health Sequence Classifier** (*Python, PyTorch, Optuna*) **Feb 2026 – Apr 2026**
Built a Bi-LSTM sequence classifier using custom Focal Loss to handle class imbalance, scaling training via mixed precision (AMP) and designing custom regex-based, zero-padded tokenization pipelines.
- **Unsupervised Movie Cluster Analysis** (*Python, Scikit-Learn*) **Mar 2026 - Mar 2026**
Designed an unsupervised machine learning pipeline utilizing agglomerative hierarchical clustering with Jaccard similarity metrics to map high-density, stable cinematic genres from raw metadata.

AI Automation Tool

- **Git Workflow Automation CLI** (*Python, Git API, GitHub CLI, Ollama, LLMs*) **Feb 2026 – Mar 2026**
Engineered a local LLM and GitHub CLI-driven automation tool for repository management—featuring AI staging and context-aware commit generation from diff streams—while building a robust stdin-streamed pipeline with UTF-8 fallback and soft-reset history recovery to eliminate shell failures.

Cybersecurity

- **Cryptographically Secure System** (*Python, Cryptography, Blockchain*) **Jan 2026 – Mar 2026**
Engineered a zero-trust, end-to-end encrypted distributed platform using SECP256k1 ECC, ephemeral ECIES transport security, and Shamir's Secret Sharing to enforce multi-party threshold consensus for secure bid decryption.

System Design

- **Resource Management System (RMS)** (*Next.js, MySQL, Prisma, Railway*) Jan 2025 – Dec 2025
Architected a multi-tenant full-stack platform optimizing relational database schemas, implementing connection pooling for heavy traffic, and deploying automated CI/CD staging pipelines via GitHub Actions.

Computer Vision

- **Smart Parking Detection System** (*Python, OpenCV, Flask*) Jan 2025 – Jun 2025
Engineered a real-time computer vision pipeline utilizing adaptive thresholding and a custom pixel-contour weighting heuristic to eliminate lighting false positives, integrated with a multi-threaded Flask web streaming live slot occupancy via an interactive ROI calibration interface.

Computer Graphics

- **Interactive 3D Escape Room Game** (*C++, OpenGL, GLUT*) Jun 2025 – Jan 2026
Engineered a first-person 3D exploration game featuring smooth FPS camera kinematics with grid-based collision mapping, dynamic spotlighting attenuation, and an orthographic HUD overlay, modularizing engine components into structural static libraries.

IoT & Embedded Systems

- **IoT Weather Station** (*ESP32, FastAPI, TimescaleDB*) Apr 2026 – Present
Engineered a self-sustaining environmental monitoring node using an ESP32 dual-core processor, implementing hardware-level power gating (MOSFETs) to achieve < 10μA deep-sleep cycles and a hybrid Wi-Fi/GSM failover network stack.
- **Medical Payload Capsule for Drone** (*Arduino, C++, Embedded Systems*) Jan 2024 – Oct 2024
Designed a ruggedized drone-mounted transit capsule using active Peltier thermoelectric cooling and sensor telemetry fusion (IMU, Barometer) to stabilize temperature, humidity, and real-time GPS coordinates via a GSM signaling array.

EXPERIENCE

- Freelance Full-Stack Developer & Graphic Designer**
Freelancer.com (u/SuperWorksRS) & Local Clients

Jan 2020 – Present
Remote / Sri Lanka
- Execute global freelance contracts focusing on high-fidelity graphic design, asset creation, and interactive React front-end interface development to bridge the gap between aesthetic branding and UI/UX optimization.
 - Engineered a high-reliability commercial platform deployed via Vercel for zero-cost serverless hosting, implementing 3rd-party Google Review API integrations and establishing custom enterprise email domain routing to minimize overhead while maximizing client business requirements and system maintainability.

- Web Development Team Member**
Hackers' Club, Department of Computer Engineering — University of Peradeniya

Jan 2024 – Present
Peradeniya, Sri Lanka
- Engineered and maintained the club's official Next.js static production website, executing manual front-end UI enhancements, content updates, and continuous asset optimization for upcoming events.
 - Managed event-driven deployment countdowns and digital infrastructure modifications, receiving an official letter of appreciation from the Senior Treasurer and President for outstanding technical contributions.

SKILLS & COMPETENCIES

- **Languages:** Python, C, C++, C#, JavaScript/TypeScript, SQL, Bash/Shell, HTML/CSS, Java
- **Frameworks & Libraries:** PyTorch, TensorFlow/Keras, Scikit-Learn, Next.js, React, Flask, FastAPI, OpenCV, OpenGL, Transformers
- **Tools, Cloud & DBs:** Git/GitHub, GitHub Actions (CI/CD), Vercel, Railway, MySQL, Prisma, Optuna, Ollama, Anaconda
- **Embedded IoT:** ESP32, ESP32-C3, Arduino, Raspberry Pi Pico, IMU/Barometer Sensors, GSM, Embedded C++, Power Gating
- **Core Expertise:** Machine/Deep Learning, NLP, Computer Vision, Cybersecurity, Cryptography, System Design, Full-Stack Dev

REFERENCES

Prof. Saluka R. Kodituwakku
Research Supervisor
Senior Professor of Statistics and Computer Science
University of Peradeniya
B.Sc. (Perad.), M.Sc. (AIT, Thailand), Ph.D. (RMIT, Australia)